

Name: _____

Date: _____

PROBABILITY, (MUTUALLY EXCLUSIVE EVENTS)

LO: I can identify mutually exclusive events and use the addition rule for probability.

Mutually exclusive events

Mutually exclusive events cannot happen at the same time. If one occurs, the other cannot occur.

Addition rule: If A and B are mutually exclusive, $P(A \text{ or } B) = P(A) + P(B)$.

Quick examples

	Rolling a 3 and rolling a 5 on one die: mutually exclusive	cannot get both
	$P(A \text{ or } B) = P(A) + P(B)$ when mutually exclusive	addition rule
	All probabilities in a sample space sum to 1	exhaustive
	Drawing a red card and a king are mutually exclusive	king of hearts is both

Key idea

Mutually exclusive events have no overlap: $P(A \text{ and } B) = 0$.
So $P(A \text{ or } B) = P(A) + P(B)$.

$$P(A \text{ or } B) = P(A) + P(B)$$

Only when A and B cannot happen together

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - identifying mutually exclusive

A die is rolled. Are 'rolling a 2' and 'rolling an even number' mutually exclusive?

Rolling a 2 IS an even number
Both can happen on the same roll
NOT mutually exclusive

Not mutually exclusive

2 is contained within the set of even numbers.

Example 2 - addition rule

$P(\text{red}) = 0.3$, $P(\text{blue}) = 0.45$. If colours are mutually exclusive, find $P(\text{red or blue})$.

Mutually exclusive: use addition rule

$$P(\text{red or blue}) = 0.3 + 0.45 = 0.75$$

= 0.75

Simply add the probabilities when events cannot overlap.

Example 3 - finding a missing probability

A spinner has $P(\text{red}) = 0.2$, $P(\text{blue}) = 0.35$, $P(\text{green}) = 0.15$. Find $P(\text{yellow})$.

$$\begin{aligned} \text{All probabilities sum to 1} \\ 0.2 + 0.35 + 0.15 + P(\text{yellow}) &= 1 \\ P(\text{yellow}) &= 1 - 0.7 = 0.3 \end{aligned}$$

= 0.3

The sample space is exhaustive so all probabilities total 1.

Example 4 - or probability from a table

A bag: 5 red, 3 blue, 2 green (total 10). Find $P(\text{red or green})$.

Red and green are mutually exclusive (a ball is one colour)
 $P(\text{red}) = \frac{5}{10}$, $P(\text{green}) = \frac{2}{10}$
 $P(\text{red or green}) = \frac{7}{10}$

= $\frac{7}{10}$

Each ball is only one colour, so colours are mutually exclusive.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying mutually exclusive

State whether each pair is mutually exclusive (Yes/ No).

- a) Rolling a 6 and rolling a 3 (one die)
- b) Picking a heart and picking a queen (cards)
- c) Getting heads and getting tails (one coin)
- d) Being left-handed and being female

2. Using the addition rule

Events are mutually exclusive. Find $P(A \text{ or } B)$.

- a) $P(A) = 0.3$, $P(B) = 0.4$
- b) $P(A) = \frac{1}{5}$, $P(B) = \frac{2}{5}$
- c) $P(A) = 0.15$, $P(B) = 0.25$
- d) $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$

3. Finding missing probabilities

All outcomes are mutually exclusive and exhaustive.

- a) $P(A) = 0.4$, $P(B) = 0.35$, $P(C) = ?$
- b) $P(X) = \frac{1}{3}$, $P(Y) = \frac{1}{4}$, $P(Z) = ?$
- c) $P(R) = 0.1$, $P(S) = 0.3$, $P(T) = 0.25$, $P(U) = ?$
- d) $P(1) = P(2) = P(3) = P(4)$. Find $P(1)$

4. Word problems

Solve each probability problem.

- a) Bag: 6 red, 4 blue, 5 green. $P(\text{red or blue})$?
- b) Die: $P(\text{less than 3 or greater than 4})$?
- c) Cards: $P(\text{ace or king})$?
- d) Spinner 1-8: $P(2 \text{ or prime})$?

5. Mixed mutually exclusive

Answer each question.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Mutually exclusive events cannot happen at the same time

☐

Correct answer: _____

Reason: _____

b) $P(A \text{ or } B) = P(A) + P(B)$ works for all events ☐

Correct answer: _____

Reason: _____

c) If events are exhaustive, their probabilities sum to 1 ☐

Correct answer: _____

Reason: _____

d) Rolling even and rolling a 6 are mutually exclusive ☐

Correct answer: _____

Reason: _____

e) Mutually exclusive events have $P(A \text{ and } B) = 0$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bag has red, blue, and green balls. $P(\text{red}) = 0.35$, $P(\text{blue}) = 0.4$. Find $P(\text{green})$. If there are **8 green balls**, how many balls are in the bag total? How many are red?

Expression: _____

Simplified: _____

b) Events A, B, C are mutually exclusive and exhaustive. $P(A) = 2P(B)$ and $P(C) = 3P(B)$. Find $P(A)$, $P(B)$ and $P(C)$. If the experiment is done **120 times**, how many times would you expect each event?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

